

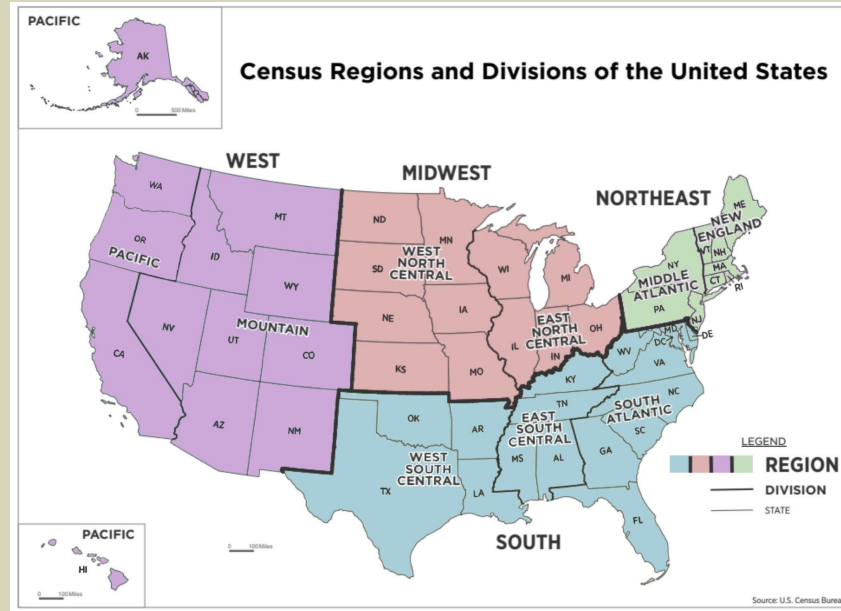
MBN Disorder ER Diagnosis Rate **by** **US Region**

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MBN Disorders

“Mental, behavioral, and neurodevelopmental disorders are syndromes characterized by clinically significant disturbance in... mental and behavioral functioning.” (World Health Organization, 2019)¹

United States Regions



(United States Census Bureau, 2021)²

Northeast Average	Midwest Average	South Average	West Average
22.13	27.17	29.42	19.91

Quality of Life State Ranking (Blanchflower & Bryson, 2024)³

Research Question & Hypothesis

RESEARCH QUESTION

How do emergency room mental, behavioral, and neurodevelopment diagnosis rates vary across regions (Northeast, Midwest, South, and West) of the United States?

HYPOTHESIS

The Southern United States will have the lowest average emergency room diagnosis rate of the four regions.

Northeast Average	Midwest Average	South Average	West Average
7.67	24.25	35.38	29.54

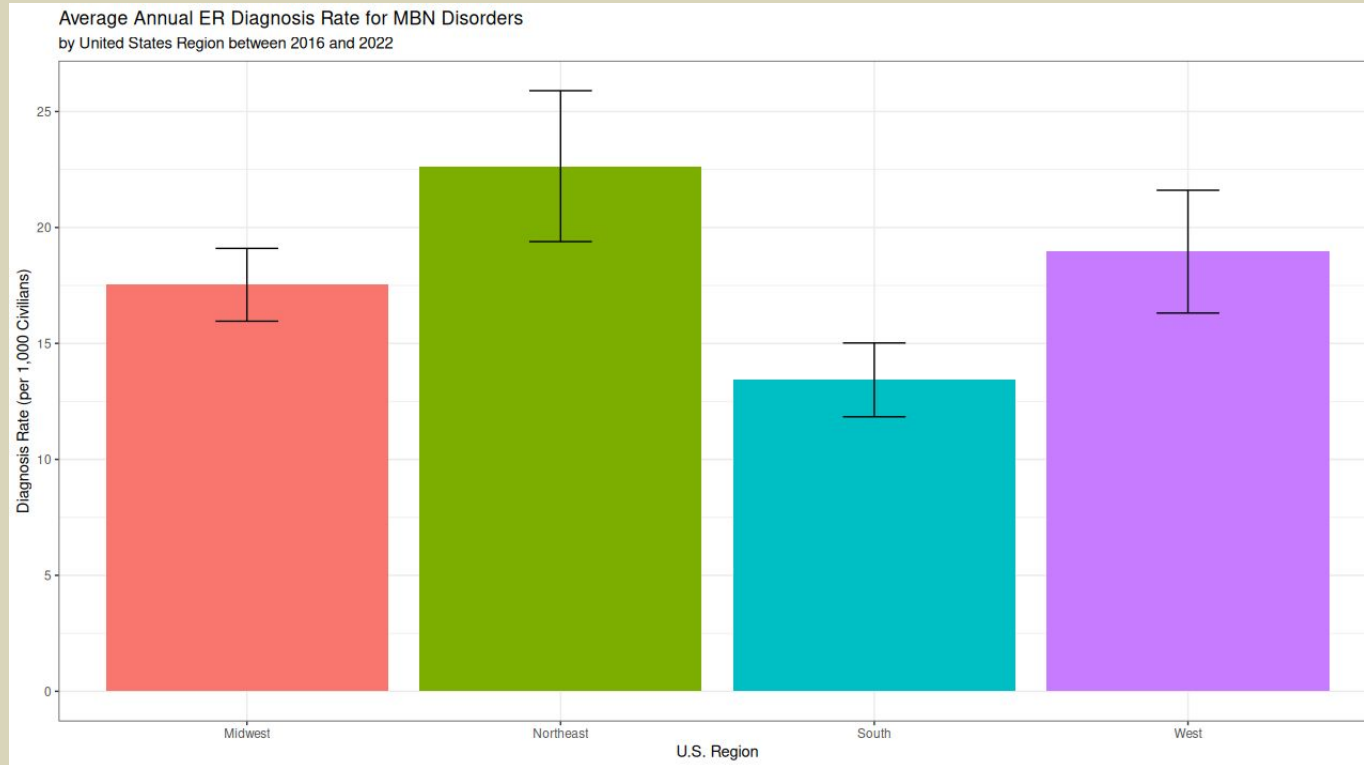
Access to Mental Healthcare State Ranking (Reinert et. al, 2025)⁴

Methods

- Clean ER visits dataset.
- Graph means.
- Run ANOVA.

Year	USRegion	Rate	SE
2016	Northeast	27.3	4.6
2016	Midwest	16.0	3.1
2016	South	12.5	1.9
2016	West	18.5	3.1
2017	Northeast	16.7	3.3
2017	Midwest	16.8	3.7
2017	South	13.1	1.8
2017	West	15.6	2.8
2018	Northeast	20.5	3.7
2018	Midwest	15.9	2.8
2018	South	12.3	1.5
2018	West	15.9	2.2
2019	Northeast	19.8	3.1
2019	Midwest	18.3	2.7
2019	South	17.4	2.4
2019	West	15.8	2.2
2020	Northeast	27.2	5.8
2020	Midwest	21.8	5.3
2020	South	10.8	1.9
2020	West	24.5	5.2
2021	Northeast	26.6	4.1
2021	Midwest	16.3	2.7
2021	South	13.2	3.2
2021	West	20.2	3.5
2022	Northeast	20.4	3.3
2022	Midwest	17.6	2.1
2022	South	14.7	3.0
2022	West	22.2	3.7

Results



Annual ER Diagnosis Rate for MBN Disorders
By United States Region between 2016 and 2022



Statistical Analysis

ANOVA p-value = 0.000154

Comparison	p-value	Significance
West vs. Northeast	p = 0.1537596	Insignificant
West vs. Midwest	p = 0.8299721	Insignificant
South vs. Midwest	p = 0.0962670	Insignificant
South vs. Northeast	p = 0.0000687	Significant
Northeast vs. Midwest	p = 0.0267767	Significant
South vs. West	p = 0.0152399	Significant

1 Discussion

ANOVA p-value =
0.000154
 $p < 0.05$

 $n = 7$

2 Conclusions

The South showed
significant
differences from
two of the other
three U.S. regions.

3 Moving Forth

Rural vs. urban
populations
rather than region
alone.

THANK YOU

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